ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A2 · CELLS

Cells and specialised cells

★☆☆
Grade 5

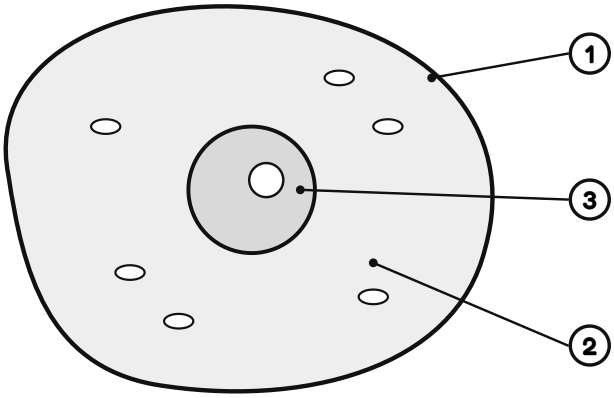
Learning goals

- Label a cell.
- Match cells to their jobs.
- Order the levels of organisation.

1 **Label the parts of the animal cell.** [3]

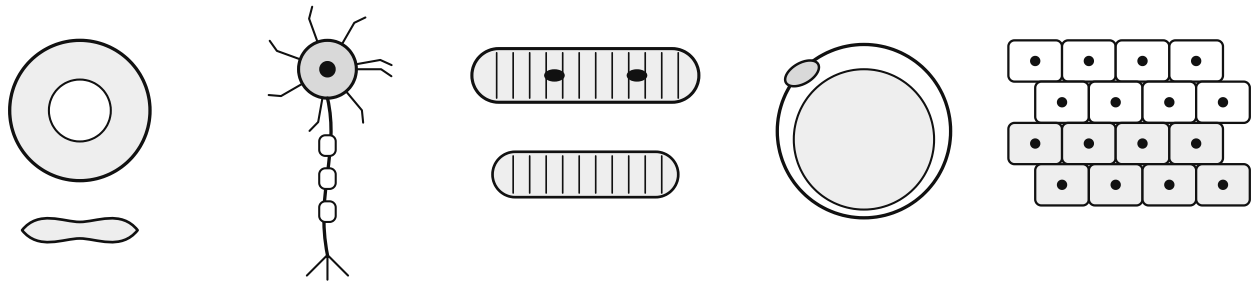
LABELS

cytoplasm **nucleus** **cell membrane**



(1) cell membrane (2) cytoplasm (3) nucleus

2 **Write the letter of each cell next to its name.** [10]



A B C D E

Cell	Letter	Its job
red blood cell	A	carries oxygen
nerve cell	B	carries messages
muscle cell	C	contracts to move the body
fat cell	D	stores energy, keeps us warm
skin cell	E	makes a protective layer

3 **Number the levels of organisation from smallest (1) to largest (5).** [2]

- [1] cell
- [3] organ
- [4] organ system
- [5] organism
- [2] tissue

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. All living things are made of cells.	✔		
2. The nucleus controls what the cell does.	✔		
3. A tissue is made of many different organs.		✔	A tissue is made of many similar cells.
4. Robert Hooke named cells after looking at cork.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A2 · CELLS

Shape and job

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Grade 5

Learning goals

- Explain how a cell's shape suits its job.
- Use the levels of organisation to explain how the body is built.

1 **Explain how the shape of each cell helps it do its job.** [6]

a) nerve cell

It is long and thin with branches, so it can carry messages a long way and connect to many other cells.

b) red blood cell

Its dented disc shape gives a large surface to carry oxygen, and it can bend to squeeze through narrow vessels.

c) skin cells

They are flat and fit tightly together, making a barrier that keeps germs out.

2 **Complete the table for the digestive system.** [5]

Level	Example
cell	a stomach muscle cell
tissue	muscle tissue in the stomach wall
organ	stomach
organ system	digestive system
organism	a human

3 **Rumana says, "If one organ stops working, the whole organism is in danger." Use the idea of organ systems to explain whether she is right.** [2]

Yes: organs depend on each other. For example, if the heart stops, oxygen from the lungs cannot reach the cells, so all the systems fail.

4 **A plant cell has a cell wall and chloroplasts. An animal cell does not. Suggest why plant cells need these.** [2]

The cell wall makes plant cells strong so the plant can stand up without bones; chloroplasts trap light to make food.

ANSWER KEY

InquiryMind
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INVESTIGATE · A2 · CELLS

Looking at cells with a microscope

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Grade 5

Learning goals

- Use a microscope safely.
- Calculate magnification.
- Observe, draw and compare cells.

You will need

microscope, a thin piece of onion skin, a glass slide, a cover slip, a drop of iodine (stain), a pencil

Stay safe

Handle glass slides carefully. Iodine stains skin and clothes. Never point a microscope mirror at the Sun.

1 **Magnification tells us how many times bigger something looks. Multiply the eyepiece by the objective lens.** [3]

Eyepiece lens	Objective lens	Total magnification
×10	×4	×40
×10	×10	×100
×10	×40	×400

2 **Follow the steps to look at onion cells. Then draw what you see.** [5]

1. Peel a thin, see-through layer from the inside of an onion piece.
2. Lay it flat on the slide and add one drop of iodine.
3. Lower a cover slip gently on top.
4. Start with the smallest objective lens, focus, then try a bigger one.

My drawing of onion cells (×100)

Brick-shaped cells in rows; a cell wall around each; a dark round nucleus in many cells.

a) Describe the shape of the cells.

Like long bricks or boxes, arranged in rows, touching each other.

b) Which part of the cell did the iodine make easier to see?

The nucleus (it stains darker).

3 **Compare onion cells with cheek cells (animal cells). Fill in the table.** [3]

Feature	Onion cell (plant)	Cheek cell (animal)
shape	regular, brick-like	irregular, rounded
cell wall?	yes	no
nucleus?	yes	yes

4 **Why do scientists add a stain such as iodine?** [1]

Cells are almost see-through; a stain colours parts of the cell so they are easier to see.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A2 · CELLS

Giving blood: cells that save lives

★★★
Grade 5

Learning goals

- Explain the jobs of the parts of blood.
- Use data and reasoning about a real health issue.
- Discuss why people choose to help others.

Science · Health · Maths · Ethics · Community

Blood looks like a red liquid, but it is really a mixture of cells floating in a yellow liquid called **plasma**. **Red blood cells** carry oxygen. **White blood cells** fight germs. Tiny **platelets** help blood clot to stop bleeding.

In Bangladesh, hospitals need blood every day for accident victims, mothers giving birth and people having operations. Many children have **thalassaemia**, an inherited condition in which the body cannot make enough healthy red blood cells. Many of these children need a blood transfusion every few weeks.

Healthy adults can give (donate) blood safely. In 1977, students at Dhaka Medical College started **Sandhani**, a volunteer group that encourages young people to give blood. Today there are many blood donor groups across the country. The body replaces the donated blood: the liquid within a day or two, and the red blood cells within a few weeks.

Information from general medical sources; donors must meet age and health rules set by their blood bank.

1 **Match each part of blood to its job.** [4]

Draw a line to match each pair, or write the letter.

red blood cells ●	→ A	● **A** carry oxygen
white blood cells ●	→ C	● **B** help blood clot
platelets ●	→ B	● **C** fight germs
plasma ●	→ D	● **D** liquid that carries the cells and nutrients

2 **A person usually gives about 450 ml of blood. An adult has about 5 litres (5000 ml) of blood.** [2]

a) How much blood is left in the donor's body straight after giving?

5000 − 450 = 4550 ml

b) About what fraction of their blood did they give? Circle: one half · one tenth · one hundredth

about one tenth (450 is about 1/10 of 5000)

3 **Why might a child with thalassaemia feel tired and weak? Use the word *oxygen*.** [2]

They do not have enough healthy red blood cells, so less oxygen reaches their cells to release energy.

4 **Only healthy adults can give blood. Suggest two reasons for this rule.** [2]

To protect the donor (children have less blood and are still growing); to make sure the blood is safe and free of infections for the patient.

5 **Why do you think thousands of students volunteer to give blood to people they will never meet? Give your opinion.** [2]

Accept thoughtful answers, e.g. kindness, it saves lives, community spirit, they may need help one day.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A2 · CELLS

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

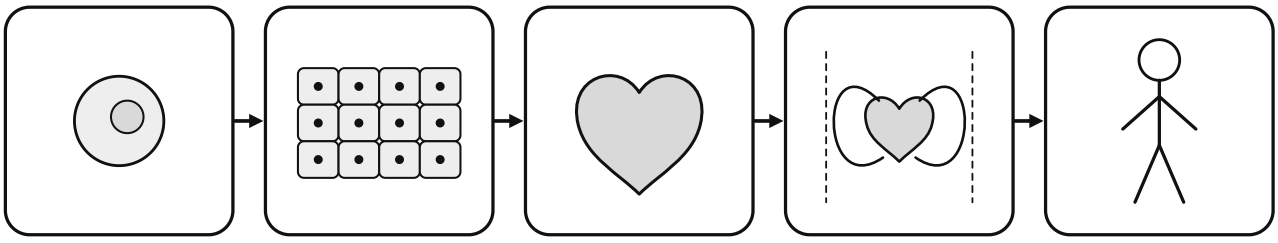
R	A	E	J	T	I	S	S	U	E	D
N	J	I	T	U	Y	C	P	R	L	F
B	A	K	K	E	P	S	S	R	P	A
H	W	G	J	O	R	G	A	N	B	T
M	S	Y	S	T	E	M	R	Y	N	F
E	I	N	W	G	I	N	T	H	E	M
L	S	K	I	N	H	T	J	N	R	U
G	V	B	L	O	O	D	L	N	V	S
B	N	U	C	L	E	U	S	K	E	C
M	O	C	C	B	U	A	V	J	G	L
O	P	C	E	L	L	S	M	K	B	E

blood **cell** **fat** **muscle** **nerve** **nucleus** **organ** **skin** **system** **tissue**

2 Label the levels of organisation. [5]

LABELS

cell **tissue** **organ system** **organ** **organism**



① cell ② tissue ③ organ

④ organ system ⑤ organism

3 Which cell carries messages around the body? [1]

- ○ red blood cell
- ● **nerve cell**
- ○ fat cell
- ○ skin cell

4 **The heart is made of muscle tissue, nerve tissue and other tissues. This makes the heart:** [1]

- ○ a cell
- ○ a tissue
- ● **an organ**
- ○ an organism

5 **Challenge. Why can we not see cells with our eyes alone, but we can see an organ like the heart?** [2]

Cells are microscopic (too small for our eyes); an organ contains millions of cells grouped together, so it is big enough to see.